

Drone-Guided Autonomous Navigation: A Cooperative UAV-AMR System for GPS-Denied Environments

Graduation Project

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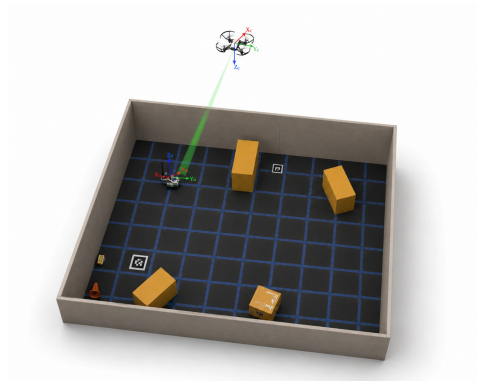
- 1 Problem & Cooperative Approach
- 2 System Architecture
- 3 Mission Pipeline
- 4 UAV Flight Control
- 5 Aerial Mapping & Image Stitching
- 6 Aerial Marker Localization
- 7 AMR State Estimation: EKF
- 8 Drive Stack, Mapping & Fusion

Setting: maritime ports, container yards, industrial sites

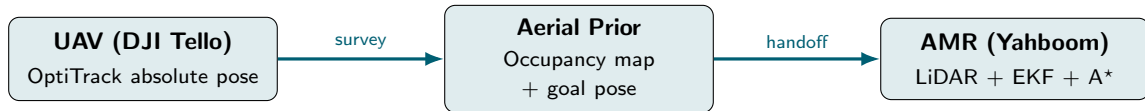
- GPS unreliable at ground level (multipath, blockage)
- No pre-built map available at mission time
- AMR carries *no* absolute-positioning sensor by design

Task: navigate to a marked goal in a 4×4 m arena

- Goal pose unknown to the ground robot a priori
- Static obstacles; no collision allowed
- Map built at runtime — no prior survey



Test arena: AMR (floor) and UAV (above)

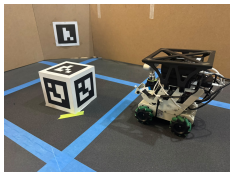


UAV role

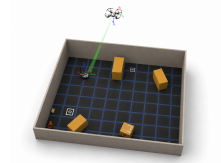
- Absolute world-frame position (OptiTrack)
- Surveys arena → builds occupancy grid
- Localizes ArUco goal markers in world frame

AMR role

- Receives aerial map + goal pose
- Fuses with live LiDAR map
- Plans A* path; feedback-linearizing controller



AMR with 3D-printed ArUco cube



UAV surveying the arena

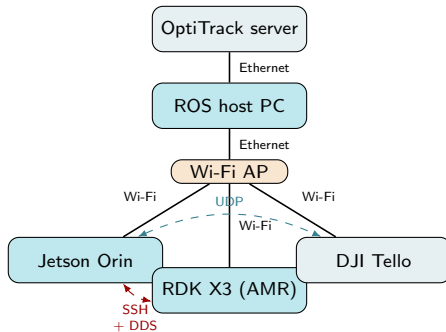
Component	Specification
AMR base	Yahboom RDK X3 (Cortex-A53)
Co-processor	Jetson Orin Nano 8 GB + GPU
LiDAR	OraDAR MS200, 360°, 15 Hz
Camera	Intel RealSense D435i
UAV	DJI Tello + 45° mirror rig
UAV positioning	OptiTrack VRPN (sub-mm)

Compute split

RDK X3: motor control, wheel odometry, EKF

Jetson Orin: stitching, planning, localization, security





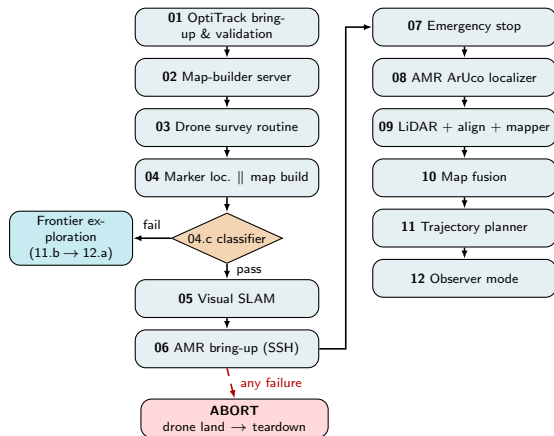
Two communication planes

Control: SSH (Paramiko) — Jetson manages RDK X3 lifecycle

Data: ROS 2 DDS over Wi-Fi
Tello bridged over proprietary UDP SDK
OptiTrack via VRPN

Latched QoS (NFR-9)

Map and goal topics use TRANSIENT_LOCAL — late-joining nodes still receive them



Key design

Stages advance on success only
Any failure → single ABORT sink
Abort: land drone first, then teardown

Stage 04.c: map-quality branch

PASS → drone map + goal pose

FAIL → border template; arm
frontier-exploration fallback

Concurrency

Map build || marker localization

Wall-clock: $\max(T_{\text{stitch}}, T_{\text{PnP}})$

World-frame PID with feedforward and anti-windup:

$$v_x^W = -k_p e_x - k_d \tilde{v}_x - k_i I_x + v_{ff,x}$$

Body-frame rotation (Tello accepts body-frame only):

$$\begin{pmatrix} v_x^B \\ v_y^B \end{pmatrix} = \begin{pmatrix} \cos \psi & \sin \psi \\ -\sin \psi & \cos \psi \end{pmatrix} \begin{pmatrix} v_x^W \\ v_y^W \end{pmatrix}$$

Survey: arc-length cubic spline, zero endpoint velocities

Fault detector: sliding-window bad-sample ratio

$\hat{\rho}_k = \frac{1}{W} \sum_j b_j > \rho_{thr}$ — handles alternating-ghost dropouts

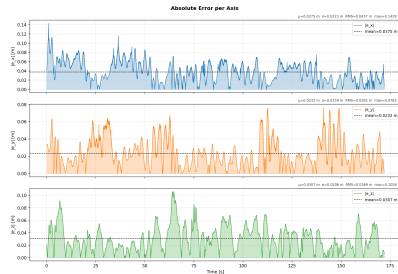
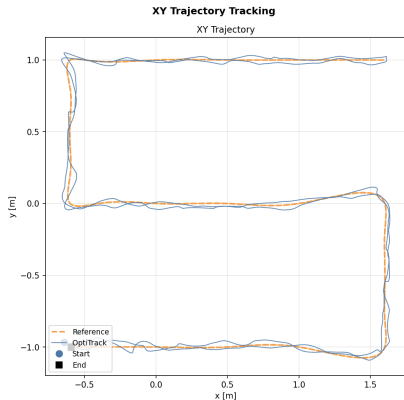
Deployed gains

Param	Value
k_p	2.0, 2.0, 3.0 (x,y,z)
k_d	0.1, 0.1, 0.3
k_i	0, 0, 0.1
v_{max}	0.6 m s ⁻¹

6-state flight FSM

pre-flight → takeoff → stabilize → **trajectory** → return → land

+ fault-land reachable from stabilize/trajectory/return



Error statistics

Axis	Mean $ e $	RMS	Peak
x	3.8 cm	4.4 cm	14.3 cm
y	2.3 cm	2.8 cm	7.6 cm
z	3.1 cm	3.7 cm	10.6 cm

Peak excursions at turn points; within stitching overlap margin

Full homography (8 DOF): over-parameterized

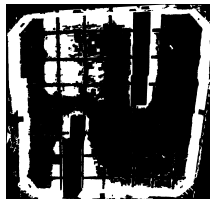
- Arena has repetitive blue-tape grid
- Extra DOF let RANSAC match tape junctions $\rightarrow$ “fan” artifacts

Similarity transform (4 DOF): translation + rotation + scale

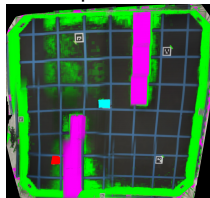
$$\mathbf{H}_{\text{sim}} = \begin{pmatrix} a & -b & t_x \\ b & a & t_y \\ 0 & 0 & 1 \end{pmatrix}, \quad a = s \cos \theta$$

- Exact model for nadir camera over a flat plane
- Stable across hundreds of frames
- `estimateAffinePartial2D` + RANSAC

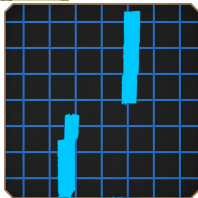
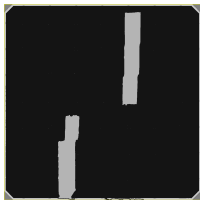
Matching: k -NN $\rightarrow$ Lowe ratio ($r=0.65$) $\rightarrow$ mutual cross-check $\rightarrow$ MAD pre-filter $\rightarrow$ RANSAC



HSV blue-tape exclusion mask



Masked incremental stitch



Final stitched arena map

Obstacle transfer on occupancy grid

Obstacle consistency scoring

$$C = \frac{w_A s_A + w_B s_B + w_C s_C}{w_A + w_B + w_C}$$

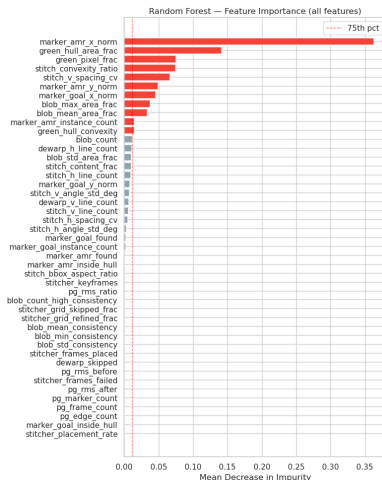
s_A : centroid agreement, s_B : area ratio, s_C : perturbation stability

w_A, w_B, w_C : fixed weights of each sub-score
= (0.45, 0.25, 0.30)

Low-confidence obstacles → wider cautionary halos

Memory budget (NFR-5)

ROI warping + 8000×8000 pre-allocated canvas
Peak working set $\approx$ 287 MB on Jetson



- 46 diagnostic features per map-build run
- Random Forest, 50-iter search, 5-fold CV; optimizes PR-AUC
- Inference: pure NumPy (no scikit-learn at runtime)

Asymmetric cost

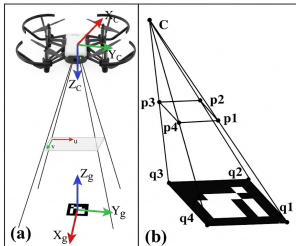
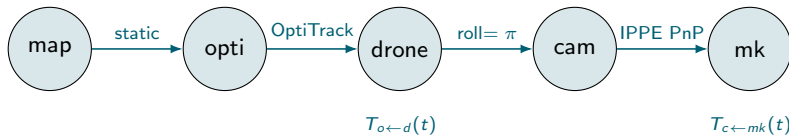
False positive (accept bad map) is worse than false negative (frontier exploration)

Threshold tuned to target precision on pass

Top features: stitching residuals, green-mask wall coverage, blob statistics — physically meaningful predictors

Random Forest feature importance (46 features)

Four-link transform chain: $T_{\text{map} \leftarrow \text{mk}} = T_{\text{m} \leftarrow \text{o}} T_{\text{o} \leftarrow \text{d}}(t) T_{\text{d} \leftarrow \text{c}} T_{\text{c} \leftarrow \text{mk}}(t)$



UAV in nadir view via 45° mirror

- **IPPE_SQUARE**: closed-form planar PnP; two-candidate ambiguity resolved by reprojection error (< 4 px)
- **Robust aggregation** (120–180 obs./marker):
MAD gate ($k=3.5$) $\rightarrow$ Weiszfeld geometric median (breakdown $\sim 50\%$) $\rightarrow$ circular median for yaw
- Position from stitcher (metric pixel); orientation from PnP (unaffected by map-origin shifts)

State: $\mathbf{X} = [x, y, \theta, v, \omega]^\top$

$$f(\mathbf{X}) = [x + T_s v \cos \theta, y + T_s v \sin \theta, \theta + T_s \omega, v, \omega]^\top$$

$$Q = \text{diag}(10^{-4}, 10^{-4}, 10^{-4}, 0.05, 0.05)$$

Three sequential updates per step:

Wheel encoder (v, ω) $R_{wh} = \text{diag}(0.2, 0.2)$

IMU gyro ω $R_{imu} = 0.001$

VIO v_x v $R_{vio} = 0.1$

Joseph-form covariance update (preserves PSD)

The VIO problem

cuVSLAM back-end *loop closures* inject velocity impulses: $|v_x^{\text{raw}}| \approx 4.9 \text{ m s}^{-1}$ vs. true cruise $\sim 0.2 \text{ m s}^{-1}$

Design choices

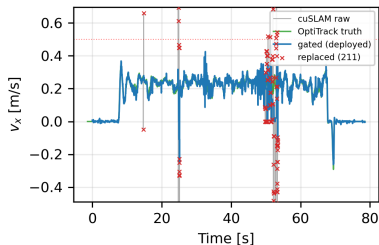
Only forward velocity *scalar* fused
VIO position discarded (drifts and jumps)
Implemented **from scratch** (NFR-1)

P0: hard clip at $|v_x| > 0.5 \text{ m s}^{-1}$

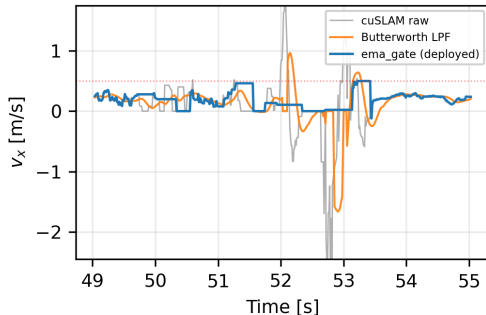
P1: EMA mean/variance tracking:

$$\hat{\mu}_k = \alpha v_k + (1 - \alpha) \hat{\mu}_{k-1}, \quad \tau_k = \max(2\hat{\sigma}_k, 0.17)$$

Burst mode: spike fraction > 0.30 in 10-sample window $\rightarrow$ hold last_good



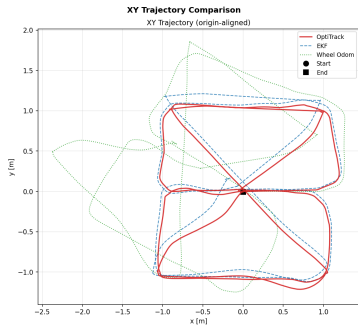
Raw cuVSLAM, OptiTrack truth, and gate output



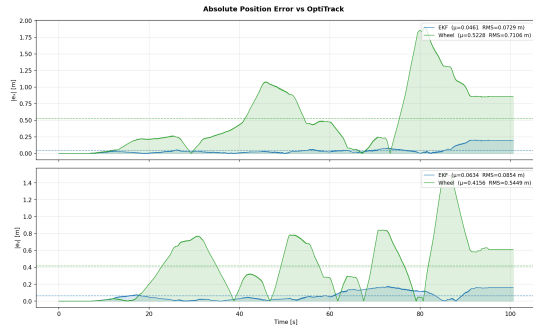
Filter ablation over a loop-closure burst

Why not Butterworth LPF?

The LPF rings in response to spikes — transient propagates a sustained deviation. `ema_gate` holds flat, close to truth.



xy trajectory: truth, EKF, wheel odometry



Absolute position error vs. OptiTrack

	EKF	Wheel only
Mean $ e_x $	4.6 cm	52.3 cm
Mean $ e_y $	6.3 cm	41.6 cm

~88% reduction
in mean position error

Controlling the robot centre is singular for the unicycle.

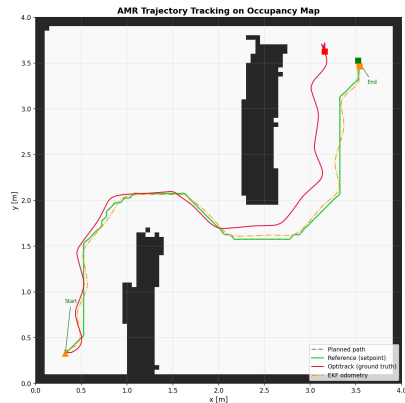
Control a **virtual point** $R = 0.2$ m ahead:

$$\begin{pmatrix} \dot{x}_R \\ \dot{y}_R \end{pmatrix} = \underbrace{\begin{pmatrix} \cos \theta & -R \sin \theta \\ \sin \theta & R \cos \theta \end{pmatrix}}_{T(\theta)} \begin{pmatrix} v \\ \omega \end{pmatrix}$$

$\det T = R \neq 0 \Rightarrow$ globally invertible

Decoupled: $\dot{e}_x = -k_{px}e_x$, $\dot{e}_y = -k_{py}e_y$

Reads world $\rightarrow$ base_footprint TF —
localization-agnostic



AMR following planned reference trajectory

Bayesian log-odds update:

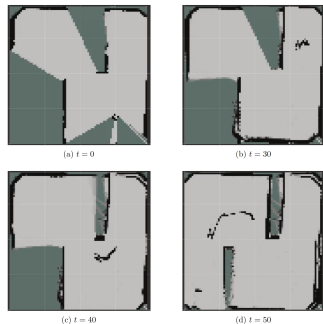
$$l_t = l_{t-1} + l_{\text{sensor}} - l_0$$

- $l_{\text{occ}} = +0.85$, $l_{\text{free}} = -0.40$ (2.1:1 asymmetry)
- Bresenham ray casting; saturated bounds
- ~ 6 hits or 13 misses to commit

Priority-weighted map fusion (no registration):

$$G_F = \begin{cases} 100 & G_D \geq 65 \\ G_A & G_D = -1 \\ \max(25, G_A) & G_D < 65 \\ 25 & \text{otherwise} \end{cases}$$

Both maps in world frame at same resolution — no registration needed



Occupancy map under construction

Fusion rule

$G_D \geq 65$: drone obstacle (immutable)

$G_D = -1$: defer to AMR map

Otherwise: AMR can only *raise* occupancy

Costmap inflation: $c(d) = 99 e^{-k(d-r_{\text{robot}})}$, $k = 3.5$

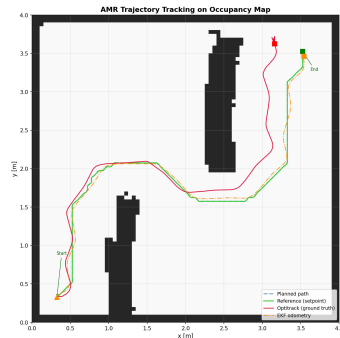
A* on 8-connected grid, octile heuristic:

$$h = \max(\Delta x, \Delta y) + (\sqrt{2} - 1) \min(\Delta x, \Delta y)$$

Replanning: goal change, path blockage, elapsed time, proximity score

Trajectory: arc-length cubic spline + **trapezoidal speed profile**

$$d_{\text{ramp}} = v^2 / 2a = 0.225 \text{ m}$$



AMR on a planned path

Replanning: < 500 ms (NFR-3)

Fallback: frontier exploration when map quality fails

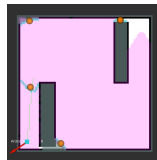
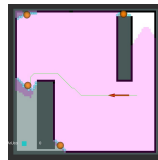
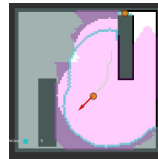
Triggered: map-quality classifier rejects drone map
(Stage 04.c FAIL)



Frontier scoring (heading-inflated distance):

$$\text{score} = w_d \widehat{\text{prox}} + w_s \widehat{\text{area}}, \quad (w_d, w_s) = (0.85, 0.15)$$

Heading penalty $\left(\frac{1+\cos \Delta \theta}{2}\right)^3$ — clusters $>60^\circ$ off-heading scaled by 0.01, suppressing U-turns



Explored (pink) expands into unknown (grey);
cyan = frontier cells; robot homes via A*

Homing

ArucoWorldBridge converts PnP detection → world-frame pose via TF → forwarded to A*. Same planner stack as main mission.

Safety nets

Stall watchdog: no 0.05 m in 4 s → re-issue goal
A* fail $\times 2$ → blacklist cluster
Detection lost → back to EXPLORING

Independent safety layer — decoupled from planning

- Blind-spot mask: $[120^\circ, 240^\circ]$ excluded
- $n=5$ consecutive scans with obstacle ≤ 0.3 m $\rightarrow$ STOP
- Clear after n clean scans
- $\tau = n/f_s = 5/15 \approx 333$ ms

Safety bound: $d_{\min} > v \cdot \tau + d_{\text{brake}}$

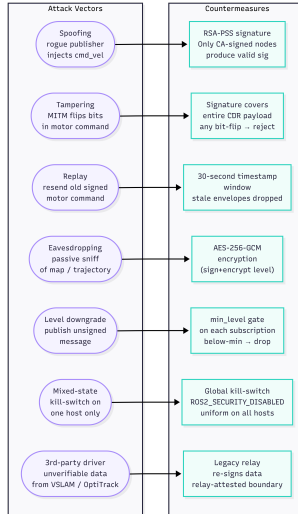
Mission check (Stage 07)

FSM verifies `/amr/emergency_stop` inactive before proceeding

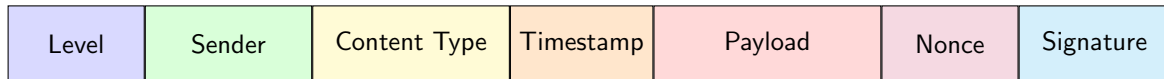
Characterization ($N=30$ trials)

Metric	Value
Mean response time	0.34 s
Stopping distance	0.18 m
True-positive rate	28/30
False triggers (20 min)	0
Recovery rate	29/30

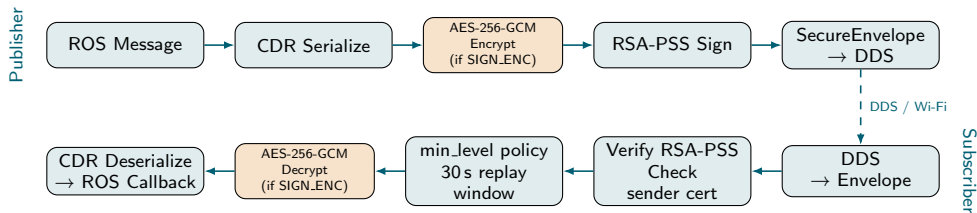
$v=0.30$ m/s; threshold 0.30 m; 15 Hz



Seven attack vectors (top) paired with countermeasures (bottom)



- **Sender** identifies the publishing node
- **Timestamp** enables replay protection
- **Payload** contains serialized ROS data
- **Nonce** used only for AES-GCM encryption
- **Signature** used for RSA-PSS authentication

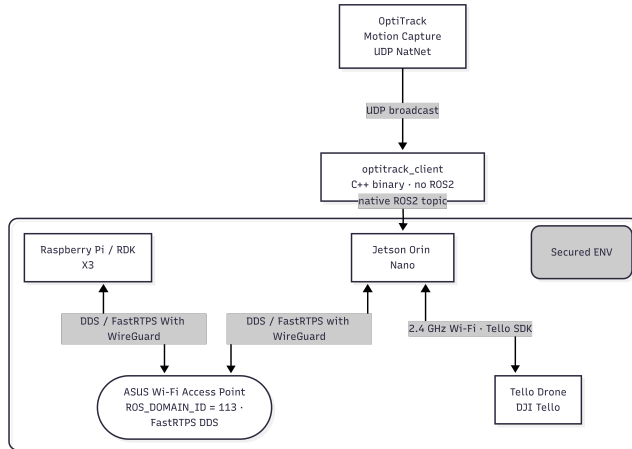


PKI

Self-signed CA (RSA-4096) issues per-node
RSA-2048 X.509 certs
AES key = SHA-256(pub-key DER) — no secret
distribution

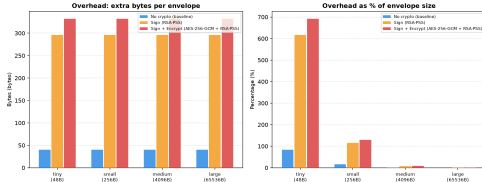
Kill switch (NFR-10)

ROS2_SECURITY_DISABLED degrades the entire
system uniformly — avoids DDS type mismatch



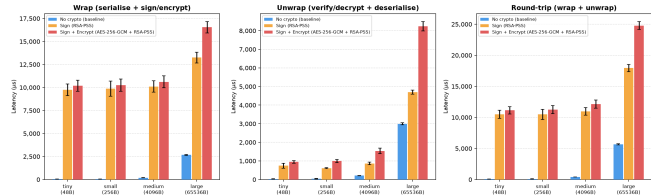
DDS/FastRTPS over WireGuard-tunnelled Wi-Fi with SecureEnvelope layer.
OptiTrack reaches the robot LAN through a native (non-secured) `optitrack_client`.

Envelope Size Overhead



Per-message cryptographic overhead by level

Crypto Latency by Security Level & Payload Size



End-to-end latency distribution

Key numbers (small payloads ≤ 4 kB)

RSA-PSS signing ≈ 10 ms per message (private-key exponentiation dominates)
marginal cost

AES-256-GCM adds only +7–11%
Applied selectively to low-frequency mission-critical topics (**NFR-11 satisfied**)

UAV flight control

- Mean tracking error ≤ 3.8 cm/axis; RMS ≤ 4.4 cm
- Within stitching overlap margin

AMR state estimation (EKF)

- Mean $|e_x|=4.6$ cm, $|e_y|=6.3$ cm vs. OptiTrack
- $\approx 88\%$ reduction vs. wheel odometry alone

Physical safety (AEB)

- 30/30 true-positive stops at 0.34 s mean response
- Zero false positives in 20 min obstacle-free traversal

Security middleware

- All 7 attack-vector suites: 100% pass
- SIGN_ENCRYPT: ~ 10 ms overhead per message
- AES adds only +7–11% over signing alone
- Kill switch: 20/20 scenarios validated

System integration

- 12-stage FSM with graceful abort
- Map-quality gate: goal-directed or frontier exploration
- Decentralized ROS 2 Humble across 3 platforms

Technical contributions:

- Similarity-transform stitching + learned map-quality gate
- From-scratch 5-state EKF with adaptive VIO outlier gate
- Robust ArUco localization: geometric median + bias calibration
- Registration-free priority-weighted map fusion
- Independent streak-detector AEB layer
- Application-layer RSA-PSS + AES-256-GCM security middleware
- 12-stage orchestrator with graceful PASS/FAIL branching

Future work:

- More frequent ArUco re-anchoring (bound dead-reckoning drift)
- Online gyro-bias estimation
- Dynamic-obstacle prediction in planner
- Scale to larger multi-robot teams
- Replace RSA sign-per-message with session keys

Open source

David-0C17/collab_nav_ground-jetson
NoeRos22/collab_nav_ground-rasp
FredRomo/security_middleware

Thank You

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